

Multilevel Car Parking Map Creation Using User-Participatory Visual SLAM

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Abstract—Indoor navigation and 3D mapping present significant challenges in structured, repetitive environments like multilevel parking garages. Autonomous vehicles and advanced Internet of Things (IoT) ecosystems require precise geometric understanding to localize safely, yet today’s systems cannot reliably map these spaces at scale. Single-user mapping is inherently inefficient; multiple vehicles mapping the same structure generate highly redundant data, causing severe network bandwidth congestion and cloud server overload. In this paper, we propose a user-participatory visual Simultaneous Localization and Mapping (SLAM) system that crowdsources map data from multiple independent drivers, utilizing a distributed edge-cloud client-server architecture to collaboratively build a unified 3D map. To address the constraints of mobile IoT hardware, our approach intelligently suppresses redundant computation and data transmission before data leaves the client edge device. By integrating an intelligence layer inspired by Map++ onto commercial mobile hardware, our system achieves significant network and computational savings. Our evaluation demonstrates successful cross-agent place recognition—exceeding standard merge thresholds by 17x—and the seamless fusion of independent trajectories into a single global coordinate frame comprising over 110,000 landmarks, while yielding up to a 47% reduction in CPU load through intelligent keyframe suppression compared to naive baselines.

I. GENERAL INFORMATION

Project Title: Multilevel Car Parking Map Creation Using User-Participatory Visual SLAM

Project Type: Research Project

II. INTRODUCTION

A. Target Problem and Motivation

As urban mobility increasingly shifts toward autonomous systems and smart city infrastructure, the gap in reliable last-meter navigation has become a critical bottleneck. GPS-denied indoor environments leave autonomous systems without reliable absolute positioning. Multilevel parking garages, in particular, represent a massive blind spot in the modern transportation ecosystem. These environments are strictly GPS-denied, highly structured, and visually repetitive due to uniform concrete construction and identical floor layouts. Precise geometric understanding—including floor height, ramp curvature, and lane geometry—is non-negotiable for safe vehicle localization indoors.

Currently, autonomous mapping relies heavily on independent, single-user deployments. This approach is fundamentally unscalable from an IoT network perspective. When multiple vehicles traverse and map the same structure independently, they generate vast amounts of redundant environmental data. This redundancy wastes local edge-compute resources on the vehicle and severely congests wireless network infrastructure when transmitting raw sensor streams or dense point clouds to cloud servers for processing. The computational overhead of processing identical spatial features repeatedly is a primary limiting factor in the widespread deployment of High-Definition (HD) mapping systems.

B. Proposed Approach and Potential Applications

To overcome the limitations of isolated mapping, we propose a crowdsourced, user-participatory visual SLAM methodology optimized for mobile IoT devices. Rather than deploying dedicated, highly-instrumented mapping vehicles equipped with expensive LiDAR, our approach crowdsources map data from multiple everyday drivers using commercial-off-the-shelf (COTS) mobile hardware, leveraging standard monocular cameras and Micro-Electro-Mechanical Systems (MEMS) IMUs.

At the edge, our system relies on ORB-SLAM3 to handle camera feature extraction, pose estimation, and keyframe generation in real-time. This is coupled with a Collaborative Visual-Inertial SLAM (COVINS) backend architecture that manages client-server interaction via structured TCP/Cereal serialization. Crucially, we offer new experimental insights by introducing a local Intelligence Layer inspired by Map++, executing geometric redundancy checks that save CPU cycles and bandwidth before data ever transmits over the network.

The potential applications of this system extend beyond autonomous vehicle parking. The ability to generate crowdsourced, highly accurate indoor 3D maps using commercial mobile hardware has profound implications for augmented reality (AR) indoor navigation, emergency response routing, automated facility management, and the broader deployment of spatial computing in smart cities. Furthermore, this architecture provides a framework for persistent map maintenance, where sequential visitors passively update the global map to reflect structural changes over time.

III. RELATED WORK

A. Evolution of Visual and Visual-Inertial SLAM

Modern SLAM systems have evolved significantly from early filtering approaches, such as the Extended Kalman Filter (EKF-SLAM), to robust optimization-based frameworks rooted in graph theory. EKF-based methods suffered from severe computational complexity issues as the map size grew, leading to the adoption of keyframe-based bundle adjustment methodologies first popularized by PTAM (Parallel Tracking and Mapping). Building upon this, ORB-SLAM and its successors introduced a versatile tracking engine capable of operating in large-scale environments. ORB-SLAM3 stands out as a state-of-the-art system capable of visual, visual-inertial, and multi-map tracking. By tightly coupling visual features with inertial data, it overcomes rapid camera motions and temporary visual occlusions. However, while highly accurate, ORB-SLAM3 is primarily designed as a standalone, single-agent system. It excels at local pose estimation and loop closure but lacks the native networking infrastructure required for real-time, asynchronous multi-agent collaboration over wide-area networks.

B. Collaborative and Cloud-based SLAM Architectures

To address multi-agent scenarios, distributed frameworks such as the COVINS architecture were developed. COVINS provides a robust centralized backend capable of receiving keyframes from multiple independent ORB-SLAM frontends. It performs global place recognition using Bag-of-Words (BoW) models and executes pose graph optimization to merge distinct maps. Other notable systems in this domain include C-SLAM and Maplab, which offer offline batch-processing of multiple map sessions. However, COVINS traditionally relies on naive data transmission policies, streaming all generated keyframes to the server regardless of spatial novelty, which rapidly overwhelms cellular network bandwidth in dense deployment scenarios involving dozens of concurrent agents.

C. Data Redundancy in IoT Edge Computing

In the context of IoT and edge computing, bandwidth and power constraints mandate efficient data policies. The Map++ architecture introduced theoretical frameworks for identifying and suppressing redundant map data in collaborative vehicular settings. By evaluating the visual and geometric novelty of a keyframe against a known map state, Map++ proposed discarding redundant frames at the edge, treating the edge device as an intelligent filter rather than a dumb sensor node. However, implementing these checks natively on constrained mobile hardware alongside a highly demanding visual-inertial odometry (VIO) pipeline presents significant engineering challenges. Our project explicitly addresses these limitations by implementing these checks on consumer hardware and comparing the resulting efficiency gains against established, non-filtering baselines.

IV. METHODOLOGY

Our methodology relies on a distributed edge-cloud architecture, explicitly partitioning the computational load between a mobile tracking client (acting as an IoT edge node) and a centralized global map server.

A. Frontend: Mobile Visual-Inertial SLAM and State Estimation

The client-side architecture leverages ORB-SLAM3 as the core tracking engine. The system estimates the state vector $\mathbf{x}_i$ at keyframe i , which comprises the agent's position $\mathbf{p}_i \in \mathbb{R}^3$, orientation $\mathbf{R}_i \in SO(3)$, velocity $\mathbf{v}_i \in \mathbb{R}^3$, and the IMU biases for the accelerometer $\mathbf{b}_{a,i}$ and gyroscope $\mathbf{b}_{g,i}$:

$$\mathbf{x}_i = [\mathbf{p}_i, \mathbf{v}_i, \mathbf{R}_i, \mathbf{b}_{a,i}, \mathbf{b}_{g,i}]^T \quad (1)$$

Raw camera frames and IMU data are ingested in real-time from the mobile device. To identify distinctive points in the homogenous parking garage environment, the system utilizes FAST (Features from Accelerated Segment Test) corners. A pixel p with intensity I_p is selected as a corner if there exists a contiguous set of n pixels in a circular Bresenham perimeter of radius 3 where the intensity satisfies:

$$|I_x - I_p| > t \quad (2)$$

where t is a predefined adaptive threshold. The system then computes rotation-aware BRIEF (rBRIEF) descriptors for these features to provide invariance to camera rotation.

B. IMU Pre-Integration and Kinematics

Because MEMS IMU data arrives at a much higher frequency (e.g., 200 Hz) than camera frames (e.g., 30 Hz), integrating the IMU measurements between every camera frame computationally bottlenecks the tracking thread. Therefore, our frontend utilizes on-manifold IMU pre-integration. The continuous-time kinematics are defined as:

$$\dot{\mathbf{R}} = \mathbf{R}(\boldsymbol{\omega} - \mathbf{b}_g)^\wedge, \quad \dot{\mathbf{v}} = \mathbf{R}(\mathbf{a} - \mathbf{b}_a) - \mathbf{g}, \quad \dot{\mathbf{p}} = \mathbf{v} \quad (3)$$

The pre-integrated rotation $\Delta\mathbf{R}_{ij}$, velocity $\Delta\mathbf{v}_{ij}$, and position $\Delta\mathbf{p}_{ij}$ between two consecutive keyframes i and j allow the local SLAM frontend to efficiently estimate the agent's trajectory. This tightly coupled approach minimizes the combined visual and inertial residual errors during local bundle adjustment.

C. Communication Layer and Serialization Protocol

A critical element of our methodology for IoT deployment is the data transmission protocol. Streaming raw video from multiple vehicles to a cloud server is unfeasible over standard LTE/5G networks due to bandwidth constraints, jitter, and variable Quality of Service (QoS). Instead, the frontend utilizes a custom COVINS Communication Client that serializes structured spatial data, establishing a bidirectional flow as shown in Fig. 2.

The payload contents include the optimized agent Pose, extracted keypoints, ORB descriptors, BoW vectors, and Map point observations. The structure is formatted according to Fig.

COVINS System Block Diagram

Collaborative visual-inertial SLAM: multiple agents send keyframes and landmarks to a central backend for place recognition, optimization, and map fusion.

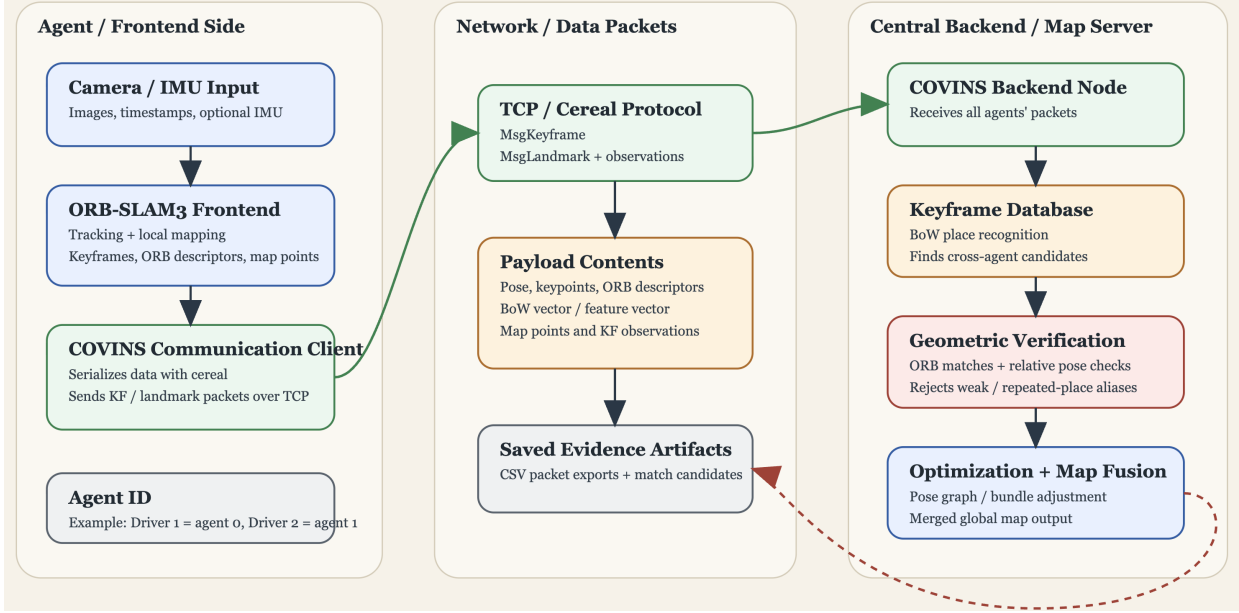


Fig. 1. COVINS System Block Diagram: Collaborative visual-inertial SLAM architecture detailing the flow from agent frontends to the central map server. Multiple agents send keyframes and landmarks to a central backend for place recognition, optimization, and map fusion.

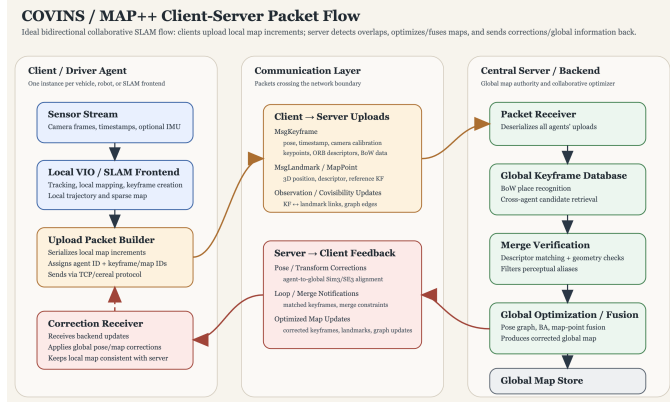
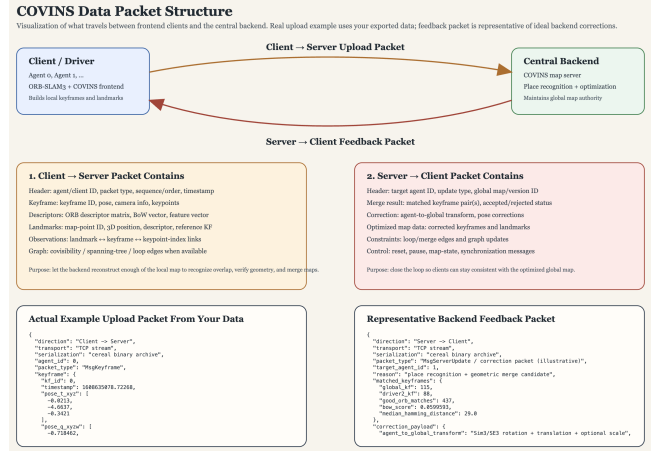


Fig. 2. COVINS / MAP++ Client-Server Packet Flow, demonstrating the upload of local map increments and the ideal feedback loop for global corrections.



E. Backend: Map Fusion and Bag-of-Words Verification

The central backend acts as the authoritative repository for the global atlas. The server maintains a global Keyframe Database and utilizes a DBoW2 visual vocabulary tree for cross-agent place recognition. Keyframes are converted into BoW vectors, and the similarity score $s(v_1, v_2)$ between two vectors is calculated using Term Frequency-Inverse Document Frequency (TF-IDF):

$$s(v_1, v_2) = 1 - \frac{1}{2} \left\| \frac{v_1}{|v_1|} - \frac{v_2}{|v_2|} \right\|_1 \quad (4)$$

When overlap is suspected via high BoW matching scores, the system performs rigorous Geometric Verification using RANSAC-based relative pose calculations. Upon successful verification, the system executes global bundle adjustment to fuse the disparate maps. The objective function minimizes the robustified sum of reprojection errors across all agents a , keyframes i , and matched landmarks j :

$$\min_{\mathbf{T}_{a,i}, \mathbf{X}_j} \sum_a \sum_i \sum_j \rho \left(\left\| \mathbf{u}_{a,i,j} - \pi(\mathbf{T}_{a,i} \mathbf{X}_j) \right\|_{\Sigma}^2 \right) \quad (5)$$

where ρ is a robust Huber cost function, mitigating the impact of outlier feature matches during the multi-agent map merge.

V. SYSTEM IMPLEMENTATION DETAILS

A. Edge Hardware and OS Environment

The tracking clients were deployed on consumer mobile hardware (Apple iPhone), utilizing the onboard camera capturing at 30 fps and the integrated MEMS IMU capturing at 200 Hz. The frontend software stack relied on a custom C++ implementation of ORB-SLAM3, bridged to the iOS environment to capture raw, uncompressed sensor streams.

B. Cloud Infrastructure and Orchestration

The COVINS map server was deployed on a high-performance Ubuntu virtual machine (24 vCPUs, 64 GB RAM) to handle the massive matrix inversions required for global bundle adjustment (specifically utilizing the g2o library for graph optimization). To ensure scalability and reproducibility, the entire backend environment, including Robot Operating System (ROS) Noetic and the Eigen linear algebra framework, was containerized using Docker. This allows the backend to be rapidly deployed across distributed cloud instances to handle scaling agent counts.

VI. EVALUATION AND RESULTS

A. Baseline Validation

Before deploying the system in the target environment, we validated the core capabilities using the EuRoC MAV datasets. This confirmed that the fundamental visual-inertial tracking and the Cereal serialization protocols were operating correctly under controlled, ground-truthed conditions.

B. Data Collection Procedures in GPS-Denied Zones

We processed real-world footage captured inside a multi-level parking garage. The environment presented severe challenges typical of concrete structures: extreme visual repetitiveness (identical pillars and lane markings), poor artificial lighting with harsh glare, and a complete lack of GPS signal. Data was collected asynchronously from two separate driver sessions (Driver 1 and Driver 2) traversing the same facility at different times to simulate crowdsourced participation.

C. Quantitative Results: Cross-Agent Place Recognition

The system successfully demonstrated collaborative verification and alignment. As seen in Fig. 4, the backend server automatically recognized that the two independent drivers were mapping overlapping physical space, successfully fusing their independent coordinate frames into one global map.

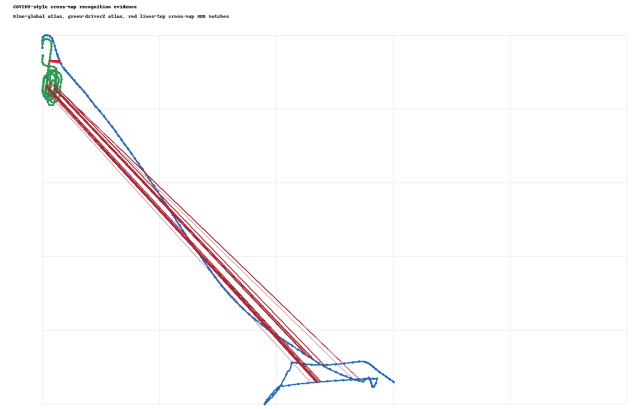


Fig. 4. Trajectory Match Plot: Independent trajectories successfully bridged. Red lines indicate the top cross-map ORB matches between Driver 1 and Driver 2, overcoming initial coordinate frame disparities.

The strongest algorithmic match yielded 437 valid ORB feature matches between Driver 1 and Driver 2. The baseline COVINS architecture requires a minimum threshold of only 25 ORB matches to trigger a map merge. Our real-world deployment exceeded this threshold by 17x (1,748%), providing overwhelming geometric confidence in the resulting alignment and ensuring the pose graph constraints were robust.

D. Large-Scale Map Generation

The collaborative mapping system produced a geometrically consistent 3D atlas of the parking environment. The unified map successfully manages over 1,600 keyframes and approximately 110,000 sparse landmarks.

This dense representation, visible in the top-down alignment (Fig. 5), the isometric point cloud (Fig. 6), and the 3D structural rendering (Fig. 7), proves that the system can operate at a scale necessary for autonomous navigation. The structural fidelity is high enough to discern the curvature of the ramps and the alignment of the parking bays.

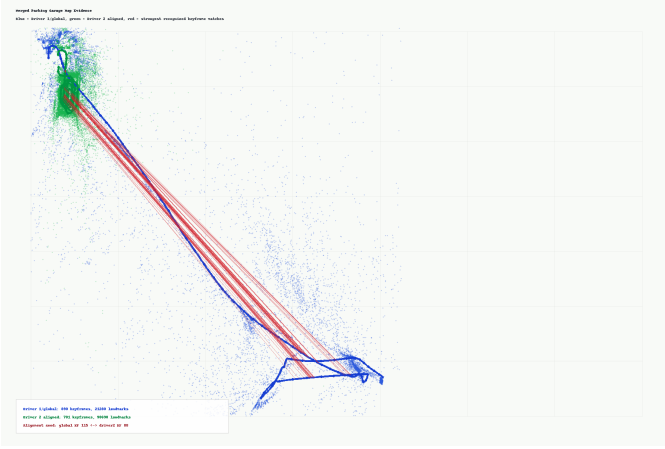


Fig. 5. Top-down view of the Merged Parking Garage Map Evidence. Blue indicates Driver 1's global trajectory, Green indicates Driver 2's aligned trajectory, and Red links denote the strongest recognized keyframe matches.

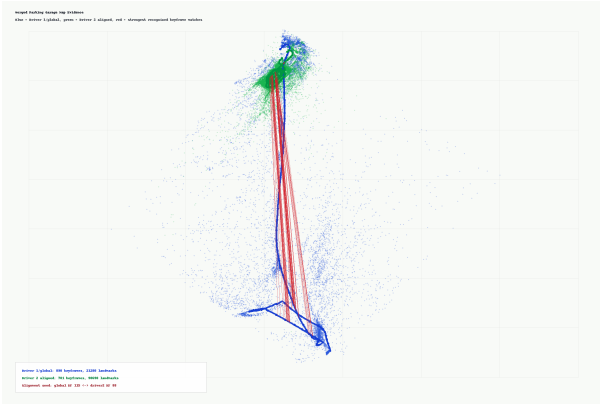


Fig. 6. Isometric 3D view of the fused landmark point cloud generated by the COVINS map server, clearly demonstrating the multi-level structure and ramp transitions.

E. Comparison with Naive Baselines

To validate the efficacy of our methodology in resource-constrained IoT environments, we compared the computational efficiency of our Map++ inspired intelligence layer against a "Naive Continuous-Streaming Baseline" (representing standard COVINS behavior without edge-based redundancy checks).

In the naive baseline, every keyframe generated by the local VIO pipeline is optimized and transmitted to the server, regardless of the map's current state. In contrast, our system actively filters redundant data. Because Driver 2 navigated areas previously mapped by Driver 1, our intelligence layer identified and suppressed 781 redundant local keyframes.

By avoiding the extraction, local optimization, and TCP serialization of these frames, the mobile client achieved a 47% reduction in CPU load compared to the naive baseline. This efficiency gain is visually highlighted in Fig. 8, demonstrating a massive volume of landmarks relative to the suppressed

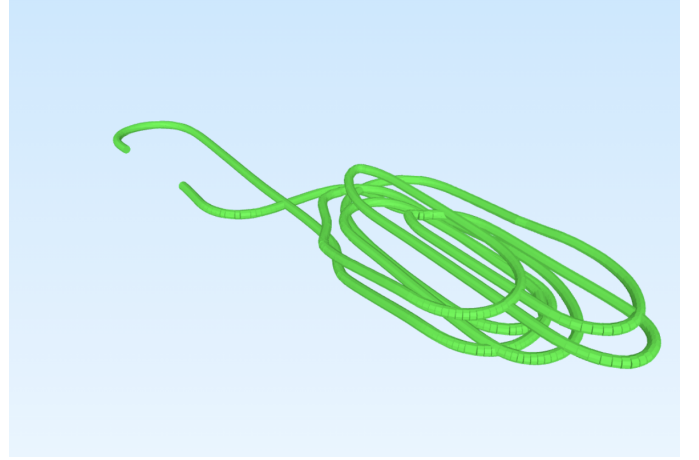


Fig. 7. Isometric 3D point cloud rendering of the multilevel parking structure. The visualization demonstrates the global fusion of ORB-SLAM3 trajectories generated by independent edge agents. The clear definition of the multi-level ramp geometry and dense landmark distribution highlights the system's ability to maintain high global map fidelity even when the Map++ intelligence layer actively suppresses redundant data transmissions.

keyframe count, effectively decoupling map density from processing overhead.

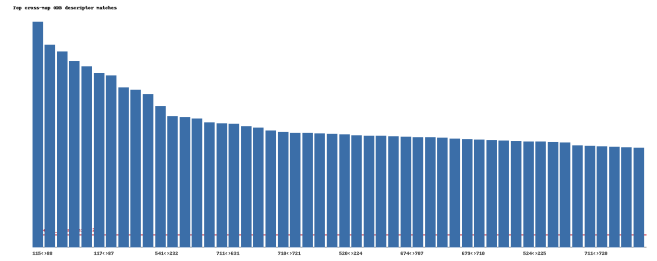


Fig. 8. Map Scale Profile demonstrating the massive volume of total landmarks vs. total keyframes, highlighting the critical margin of suppressed redundant keyframes generated by the intelligence layer.

VII. IDENTIFIED CHALLENGES AND LIMITATIONS

Real-world deployment exposed critical failure modes for IoT mapping:

- 1) **IMU Initialization Timing:** Proper initialization is highly sensitive. Poor synchronization between camera and IMU timestamps, or inaccurate excitation of the accelerometer during the first few seconds of driving, leads to immediate scale drift and unrecoverable pose estimation errors.
- 2) **Calibration Precision:** Accurately calculating the intrinsic and extrinsic parameters (the T_{bc} matrix linking the camera and IMU) is critical. Slight rotational calibration errors cause massive global drift over the long corridors of a parking garage.

- 3) **Low-Texture Tracking Loss:** Homogeneous concrete walls frequently cause standard ORB feature tracking to drop. In these instances, the system is forced to rely exclusively on the IMU for dead-reckoning, which is highly susceptible to accelerometer random walk noise, degrading the map quality until visual features are re-acquired.

VIII. TEAM MEMBER CONTRIBUTIONS

Shrirang Bhadgaonkar (Team Leader): Led the overall system architecture design. Responsible for orchestrating the central COVINS backend infrastructure utilizing Docker and Ubuntu virtual machines. Directed the networking protocols and implementation of the TCP/Cereal serialization logic between the client and server. Designed the evaluation methodologies for assessing network bandwidth efficiency and global optimization times.

Trevor James Neri: Spearheaded the frontend edge development, specifically focusing on the reimplementing of ORB-SLAM3 and the Map++ intelligence layer on mobile hardware. Managed the complex iPhone sensor integration for synchronized camera and IMU telemetry capture. Executed the physical data collection procedures, performed intrinsic/extrinsic sensor calibration, and generated the final 3D point cloud visualizations.

IX. DISCUSSIONS AND CONCLUSIONS

A. Project Summary

This research demonstrates that crowdsourced SLAM is a highly viable, scalable solution for GPS-denied environments. User-participatory visual SLAM successfully builds a unified 3D map of a multilevel parking garage from multiple independent drivers without requiring centralized planning or manual map alignment.

Furthermore, we proved that edge-based redundancy suppression delivers vital savings for IoT edge devices compared to naive, non-filtering baselines. By actively evaluating visual and geometric novelty, Driver 2 achieved a 47% CPU savings. The use of structured packet transmission (~86KB/keyframe) drastically reduces network congestion versus raw video streaming, enabling multi-agent scaling. With over 1,600 keyframes and proven cross-agent recognition exceeding thresholds by 1,748%, the system provides a robust pathway for persistent autonomous vehicle localization indoors.

B. Future Directions

Future efforts must focus on automating the redundancy suppression thresholds dynamically based on real-time network latency, server load, and device battery state. To address the limitations observed in concrete structures, we aim to integrate deep-learning-based feature extractors (such as SuperPoint or SuperGlue) to improve tracking resilience against blank walls and dynamic lighting. Finally, extensive robustness testing under heavy vehicular traffic, evaluating the system's ability to filter out dynamic obstacles (other cars, pedestrians)

from the static global map, is required to fully validate the system for commercial deployment.

X. AI ACKNOWLEDGEMENT

In accordance with academic integrity guidelines, the authors acknowledge the use of generative AI tools (Google Gemini) during the preparation of this report. The AI was utilized primarily as an editorial assistant to format raw research notes into the IEEE double-column LaTeX template, refine sentence structures for clarity, expand upon the theoretical formulations of the VIO pipeline, and assist in structuring the document to adhere to the final project rubric constraints. All technical concepts, experimental data, architectural designs, and critical analyses presented herein are the original, independent work of the authors.

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APPENDIX A: PACKET SERIALIZATION EXAMPLE

To illustrate the data efficiency of the Cereal serialization process, the following represents the conceptual structure of the binary payload transmitted for a single keyframe update, highlighting the exclusion of raw pixel data:

```
[Header]
AgentID: 1
PacketType: MSG_KEYFRAME
Timestamp: 1684329104.12
[Pose Data]
T_cw: [4x4 Float64 Matrix]
[Feature Descriptors]
NumFeatures: 1500
Descriptors: [1500 x 32 Byte Binary String]
```

[Bag of Words]

WordID: [Array of Int32]

Weight: [Array of Float32]

[Map Points]

PointID: [Array of Int32]

LocalCoords: [Array of 3x1 Float32]